

Research Notes on Rectifier neural networks

Rectifier neural networks

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

>> Section 1

$$f(x) = \ln \frac{1 + e^{kx}}{2}, f'(x) = \frac{e^{kx}}{1 + e^{kx}} = \frac{1}{1 + e^{-kx}}$$

>> Section 2

$$\log_2 \frac{1 + 2^y}{2} \approx \begin{cases} 1, & y = 0 \\ y - 1, & y \neq 0 \end{cases}$$

>> Section 3

where $b \geq 0$ is a hyperparameter that determines the "size" of the curved region near $x = 0$. (For example, letting $b = 0$ yields ReLU, and letting $b = 4$ yields the metallic mean function.) Squareplus shares many properties with softplus: It is monotonic, strictly positive, approaches 0 as $x \rightarrow -\infty$, approaches the identity as $x \rightarrow +\infty$, and is C^∞ smooth. However, squareplus can be computed using only algebraic functions, making it well-suited for settings where computational resources or instruction sets are limited. Additionally, squareplus requires no special consideration to ensure numerical stability when x is large.

>> Section 4

Piecewise-linear variants Leaky ReLU (2014) allows a small, positive gradient when the unit is inactive, helping to mitigate the vanishing gradient problem. This gradient is defined by a parameter α , typically set to 0.01–0.3.

>> Section 5

In these formulas, a , b and x_c are hyperparameter values which could be set as default constraints $a = 1$, $b = 2$ and $x_c = 1.25643$, as done in the original work.

>> Section 6

where x is the input to a neuron. This is analogous to half-wave rectification in electrical engineering. ReLU is one of the most popular activation

functions for artificial neural networks, and finds application in computer vision and speech recognition using deep neural nets and computational neuroscience.

>> Section 7

The second, less-precise approximation uses the sigmoid (logistic) function as $f(x) \approx x \cdot \text{sigmoid}(\frac{x}{1.702})$, whose formula resembles SiLU (see below).

>> Section 8

$$f(x) = \begin{cases} x & x > x_c \\ (e^{ax} - 1) / b & x \leq x_c \end{cases} f'(x) = \begin{cases} 1 & x > x_c \\ (a/b)e^{ax} & x \leq x_c \end{cases}$$

>> Section 9

and its gradient is the softmax; the softmax with the first argument set to zero is the multivariable generalization of the logistic function. Both LogSumExp and softmax are used in machine learning.

>> Section 10

$$\text{ReLU}(x) = x + \max(0, x) = x + |x|/2 = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

>> Section 11

$$f(x) = \begin{cases} x & x > 0 \\ \alpha x & x \leq 0 \end{cases} f'(x) = \begin{cases} 1 & x > 0 \\ \alpha & x \leq 0 \end{cases}$$

>> Section 12

Sparse activation: for example, in a randomly initialized network, only about 50% of hidden units are activated (i.e. have a non-zero output). Better gradient propagation: fewer vanishing gradient problems compared to sigmoidal activation functions that saturate in both directions. Efficiency: only requires comparison and addition. Scale-invariant (homogeneous, or "intensity equivariance"):

>> Section 13

$$\text{LSE}(x_1, \dots, x_n) = \ln(e^{x_1} + \dots + e^{x_n})$$

>> Section 14

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where $\tanh(x)$ is the hyperbolic tangent, and $\operatorname{softplus}(x)$ is the softplus function. Mish was obtained by experimenting with functions similar to Swish (SiLU, see above). It is non-monotonic (has a "bump") like Swish. The main new feature is that it exhibits a "self-regularizing" behavior attributed to a term in its first derivative.

>> Section 15

Parametric ReLU (PReLU, 2016) takes this idea further by making α a learnable parameter along with the other network parameters. Note that for $\alpha \leq 1$, this is equivalent to

>> Section 16

DELU Extended Exponential Linear Unit (DELU, 2023) is an activation function which is smoother within the neighborhood of zero and sharper for bigger values, allowing better allocation of neurons in the learning process for higher performance. Thanks to its unique design, it has been shown that DELU may obtain higher classification accuracy than ReLU and ELU.

>> Section 17

$f(x) = x \Phi(x) = x \cdot \frac{1}{2} [1 + \operatorname{erf}(x/\sqrt{2})]$,
 $f(x) = x \Phi(x) = x \cdot \frac{1}{2} [1 + \operatorname{erf}(x/\sqrt{2})]$

>> Section 18

$f(x) \approx \frac{1}{2} x (1 + \tanh(\frac{2}{\pi} (x + 0.044715 x^3)))$
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>> Section 19

The derivative of softplus is the logistic function. This in turn can be viewed as a smooth approximation of the derivative of the rectifier, the Heaviside step function. The multivariable generalization of single-variable softplus is the LogSumExp with the first argument set to zero:

>> Section 20

$\ln(1 + e^x) \approx \begin{cases} \ln 2, & x = 0 \\ x - e^{-x} / \ln 2, & x \neq 0 \end{cases}$
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>> Section 21

which is called the softplus (2000) or SmoothReLU function. For large negative x it is roughly $\ln 1$, so just above 0, while for large positive x it is roughly $\ln(e^x)$, so just above x . This function can be approximated as: